

Executive Summary

Milestone 2 of the TikTok Claims Classification Project

ISSUE / PROBLEM

The TikTok data team is building a machine learning model to automate the classification of claims in user submissions. As an initial step, this phase focuses on organizing the raw data and structuring it.

RESPONSE

The data team performed a preliminary investigation to identify key relationships across variables. To align with the goal of classifying user submissions, the team analyzed the relative counts of claims and opinions to evaluate the balance of video types in the dataset.

IMPACT

The discovery of null values and extreme engagement outliers means EDA must include a rigorous data-cleaning pipeline. We will use EDA to determine whether to impute or drop missing rows and decide on an outlier mitigation strategy

UNDERSTANDING THE DATA

- About 50.3% of the data are claims and 49.7% are opinions

claim_status	author_ban_status	likes_per_view			comments_per_view			shares_per_view		
		count	mean	median	count	mean	median	count	mean	median
claim	active	8596	0.329542	0.326538	5596	0.001393	0.000776	9596	0.065496	0.049279
	banned	1439	0.345871	0.358869	1439	0.001377	0.000746	1439	0.067893	0.051606
	under review	1803	0.327987	0.326867	1803	0.001367	0.000789	1803	0.065733	0.049967
opinion	active	8817	0.218744	0.218338	8817	0.000517	0.000252	8817	0.043729	0.032405
	banned	196	0.296868	0.196483	196	0.000434	0.000193	196	0.048531	0.036728
	under review	463	0.228394	0.228051	463	0.000536	0.000293	463	0.044472	0.035827

- Videos classified as "claims" accumulate exponentially higher view, share, and like counts compared to "opinions." Engineered engagement rates show clear variance, with claims maintaining an approximate 33% like-per-view ratio compared to roughly 22% for opinions.
- There is a heavy concentration of "banned" or "under review" authors tied directly to claim videos, whereas active, in-good-standing authors are more frequently associated with opinion videos

the mean of Claims is 501,029.45
the median of Claims is 501,555.00

the mean of opinions is 4,956.43,
the median of opinions is 4,953.00

KEY INSIGHTS

- Content classified as an objective "Claim" drives significantly higher view counts, shares, and downstream engagement than content classified as a subjective "Opinion".
- Authors whose account status is flagged as "Banned" or "Under Review" show a massive statistical probability of having published a claim rather than an opinion.

	claim_status	author_ban_status	video_share_count
0	claim	active	17774.5
1	claim	banned	19018.0
2	claim	under review	18084.0
3	opinion	active	121.0
4	opinion	banned	108.5
5	opinion	under review	124.0